

Tuong Luu

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EDUCATION

University of Waterloo

Candidate for Bachelor of Honours Computational Mathematics

Waterloo, Ontario

Sep 2023 - May 2028

TECHNICAL SKILLS

Languages: Java, Python, FastAPI, C/C++, JavaScript/TypeScript, React, Next.js

Databases: PostgreSQL, MongoDB, Snowflake

Tools: Git, Docker, Kubernetes (K8s), Terraform, Linux, Bash, Postman

Certifications: Azure Fundamentals, Azure AI Fundamentals

EXPERIENCE

Software Engineer Intern

Carta

Kitchener, Ontario

May. 2026 – Present

- Shipped self-service Snowflake token authentication end-to-end across Terraform, Django, and React, replacing a manual per-customer provisioning process with a feature-flagged rollout
- Migrated Carta's public API documentation to a code-first **Docusaurus** site, building a manifest-driven content pipeline and Kong/Helm routing to replace a third-party platform, saving **\$50K** a year
- Eliminated recurring on-call pages by root-causing production test failures to upstream data gaps and API gateway path misconfiguration, cutting alert noise from **255K+** to **2.5K** flagged rows
- Built **30+** regression tests for AI agent skills in an internal LLM evaluation framework, patching a framework-level matcher bug that caused intermittent false failures

Software Developer Intern

Lynkr Inc

Toronto, Ontario

Sep. 2025 – Dec. 2025

- Collaborated with engineers to develop a FastAPI and Next.js application with AI agent orchestration, generating **\$100K+ CAD** in revenue and supporting **500+ beta users**
- Migrated the database seeder to a Kubernetes job, reducing deployment time on GKE by **40%**
- Architected multi-organization support with tenant isolation and RBAC, enabling secure segregation for and independent configuration per tenant
- Built account security features with MFA using TOTP and Twilio, along with JWT session controls, cutting unauthorized access by **70%** and protecting **500+** active sessions
- Developed an audit logging system tracking all user actions, admin changes, and security events, supporting compliance for **100% of critical operations**

PROJECTS

AniDis - Full-Stack Anime Platform | Next.js, React, TypeScript, FastAPI, PostgreSQL, Redis, Docker, OpenTofu

- Designed FastAPI/PostgreSQL APIs for anime discovery, profiles, folders, episode progress, follows, threads, nested comments, and likes.
- Built an auth stack with password credentials, email verification, OTP challenges, password reset, and OAuth account linking, hardened by signed request paths and split read/write rate limits.
- Architected an **event-driven** pipeline with a transactional-outbox publisher feeding decoupled notification and email-rendering services, powering in-app reply alerts and email delivery.
- Managed the **anidis.moe** domain, DNS, and Cloudflare access policies as code in OpenTofu, running Docker Compose stacks behind Traefik with OpenTelemetry and Grafana dashboards.
- Automated releases with changesets and SHA-tagged images, plus on-demand preview environments with isolated databases.

Kyzen - 2D Game Engine | Java, LWJGL, Maven, OpenGL

- Developed a 2D game engine with batch rendering, optimizing the rendering of multiple objects in a single batch for improved performance by up to **60%** when handling **1000+** objects per frame
- Implemented an Entity-Component System (ECS), providing a modular and extendable architecture for flexible game object composition
- Integrated texture atlas support, enabling efficient texture management for sprites and tiles, reducing draw calls by over **70%**